

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 23: Graphical Representation (Histograms and Ogives)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Mode from a histogram

1. Draw a histogram for the marks of 164 students given below, and use it to find the mode.

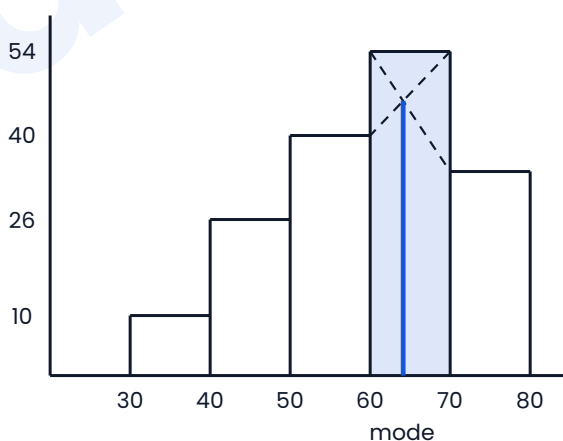
ICSE 2025, 3 marks

Marks	30–40	40–50	50–60	60–70	70–80
Number of students	10	26	40	54	34

SOLUTION

Draw a rectangle on each class with height equal to its frequency. The tallest rectangle (60–70) is the modal class.

Join the top corners of the tallest rectangle to the nearer top corners of the rectangles on either side, as shown. From the point where the two lines cross, drop a perpendicular to the marks axis.



The perpendicular meets the axis at about 64.

Check with the formula: $\text{mode} = 60 + \frac{54 - 40}{2(54) - 40 - 34} \times 10 = 60 + \frac{140}{34} \approx 64.1$

Answer: mode $\approx$ 64 marks

2. A histogram shows the heights of the plants in a school garden. Its rectangles stand on the classes 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and 90–100 (in cm) and have heights 4, 2, 8, 12, 6, 3 and 4. (a) Make the frequency table. (b) State the modal class. (c) Find the mode. (d) How many plants are between 80 cm and 90 cm tall? **ICSE 2024, 5 marks**

SOLUTION

(a) Read the height of each rectangle:

Height (cm)	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Number of plants	4	2	8	12	6	3	4

(b) The tallest rectangle is on 60–70. **Modal class: 60–70.**

(c) Join the top corners of the tallest rectangle to the nearer top corners of its neighbours.

The lines cross above 64 on the axis. Check: $60 + \frac{12 - 8}{24 - 8 - 6} \times 10 = 64$. **Mode = 64 cm.**

(d) The rectangle on 80–90 has height 3. **3 plants.**

2 Reading an ogive

3. An ogive for the monthly salaries of the workers in a factory passes through the points below. Find (a) the total number of workers (b) the median class (c) the lower-quartile class (d) the number of workers earning at least ₹6,000 but less than ₹10,000. **ICSE 2025, 4 marks**

Salary less than (₹)	2000	4000	6000	8000	10000	12000
Cumulative number of workers	10	20	35	55	70	75

SOLUTION

(a) The last cumulative frequency is the total. **75 workers.**

(b) $\frac{N}{2} = 37.5$. The cumulative frequency passes 37.5 between 6000 (35) and 8000 (55).

Median class: 6000–8000.

(c) $\frac{N}{4} = 18.75$. The cumulative frequency passes 18.75 between 2000 (10) and 4000 (20).

Lower-quartile class: 2000–4000.

(d) Workers below 10000 minus workers below 6000 = 70 – 35. **35 workers.**

4. The ages of 100 policy holders are given below. Draw an ogive and use it to find (a) the median age (b) the number of policy holders older than 52 years. **ICSE 2024, 4 marks**

Age (years)	20– 25	25– 30	30– 35	35– 40	40– 45	45– 50	50– 55	55– 60
Policy holders	2	4	12	20	28	22	8	4
Cumulative frequency	2	6	18	38	66	88	96	100

SOLUTION

Plot the points (25, 2), (30, 6), (35, 18), (40, 38), (45, 66), (50, 88), (55, 96), (60, 100), start the curve from (20, 0) and join them with a smooth curve.

(a) $\frac{N}{2} = 50$. From 50 on the vertical axis go across to the curve and down to the age axis: about 42 years.

(b) From 52 on the age axis go up to the curve and across: about 91 policy holders are 52 or younger. So about $100 - 91 = 9$ are older.

Answer: (a) median $\approx$ 42 years (b) about 9 policy holders (readings from a graph can differ by one unit)

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